DATABASE MANAGEMENT SYSTEM — PBL

Cinema Screening and Ticket Booking System

A relational database project for managing cinemas, screens, movies, screenings, bookings, tickets, concessions and payments.

by

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WHAT IS THE PROJECT?

Introduction

The Cinema Screening and Ticket Booking System is a database-based system designed to manage the complete movie ticketing process. It helps cinema staff manage screenings, seat bookings, concession orders and payments in an organized, centralized manner.

Reviewed weekly against the PBL deliverable checklist for Week 7 and Week 11 milestones.

THE SYSTEM MAINTAINS INFORMATION ABOUT

- **Cinemas & Screens**
- **Movies & Screenings**
- **Customers**
- **Bookings & Booked Seats**
- **Tickets**
- **Concession Items & Orders**
- **Payments**
- **Cancellations & Refunds**

EXISTING PROBLEM

Problem Identification

In many cinemas, ticket booking and screening information is managed using spreadsheets, manual registers or separate systems. This can lead to:

- Difficulty managing large numbers of bookings
- Duplicate or incorrect seat allocations
- Difficulty checking seat availability in real time
- Difficulty tracking cancellations and refunds
- Difficulty maintaining movie and screening schedules
- Difficulty tracking payments and concession orders
- Time-consuming report generation

PROBLEM STATEMENT

To develop a centralized database management system that efficiently manages cinemas, screens, movies, screenings, customers, bookings, tickets, concessions, cancellations, refunds and payments.

The project specification also highlights that manual seat allocation can result in the same seat being booked more than once.

Objectives

1. Maintain a centralized database of cinemas and screens
2. Manage movie and screening information
3. Track real-time seat availability
4. Record customer bookings accurately
5. Manage booked seats and generate tickets
6. Track concession item orders
7. Handle booking cancellations
8. Process refunds for cancelled bookings
9. Record and track ticket and concession payments
10. Generate useful booking and revenue reports

THE SYSTEM WILL COVER

Scope of the Project

1

Cinema Management

- Cinema details
- Screen details
- Seat layout

2

Movie Management

- Movie details
- Screening schedule
- Language & genre

3

Booking Management

- Customer details
- Bookings
- Booked seats
- Tickets

4

Payments & Concessions

- Concession items
- Orders
- Payments
- Cancellations & refunds

EXISTING APPROACHES VS PROPOSED SYSTEM

Literature Survey

Existing Approach	Limitations
Manual box office registers	Time-consuming
Excel spreadsheets	Data duplication and inconsistency
Separate booking / payment records	Difficult to integrate
Manual seat allocation	Possibility of double booking
Manual report generation	Takes more time

PROPOSED SYSTEM PROVIDES

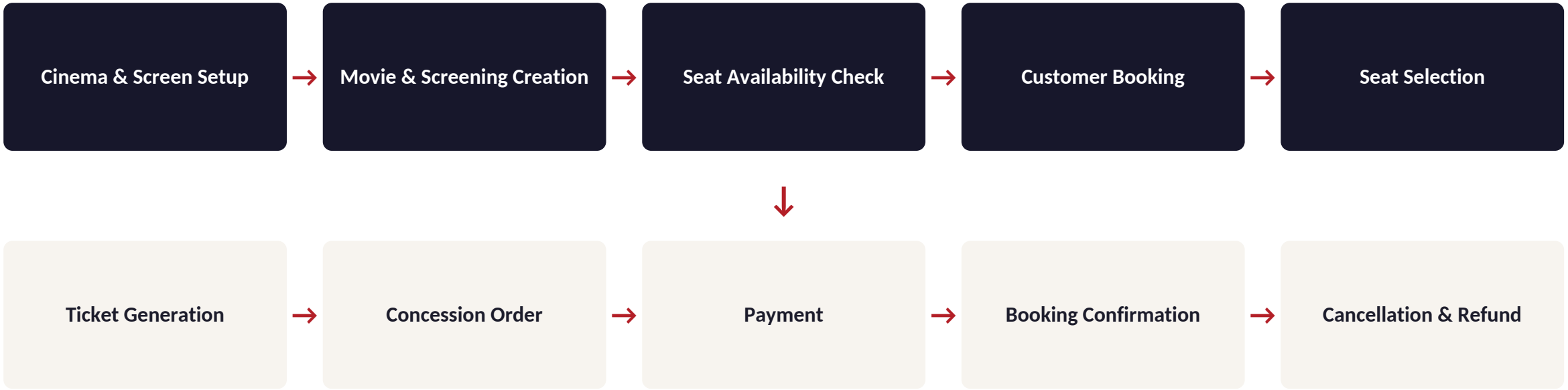
- Centralized data
- Automated seat availability
- Structured relationships
- Faster searching
- Better data integrity
- Automated reports

CONCLUSION

A centralized relational database offers better organization, consistency and retrieval than manual or spreadsheet-based booking records.

SYSTEM WORKFLOW

Proposed System



This slide explains the entire project in one flow — from cinema and screen setup through to payment and refund handling — and is well suited for verbal explanation during the review.

THREE TYPES OF USERS

Users of the System

1 Cinema Administrator

- Manage cinemas & screens
- Create movies & screenings
- Define seat layout
- Manage concession items
- Generate reports
- View all bookings

2 Customer

- View available screenings
- Check seat availability
- Book tickets
- Order concessions
- Make payments
- Cancel bookings

3 Box Office Staff

- Assist walk-in bookings
- Verify tickets
- Process payments
- Handle cancellations
- Handle refunds
- View screening schedule

SRS — THE SYSTEM SHOULD ALLOW

Functional Requirements

FR1	Admin can add, update and delete cinema and screen information.	**FR2**	Admin can add and manage movie and screening information.
FR3	System can maintain seat layout for each screen.	**FR4**	System can check real-time seat availability.
FR5	Customers can book available seats for a screening.	**FR6**	System can generate tickets for booked seats.
FR7	Customers can order concession items with a booking.	**FR8**	System can record payments for bookings and orders.
FR9	Customers can cancel a booking before the screening.	**FR10**	System can process refunds for cancelled bookings.
FR11	Users can search movies, screenings and bookings.	**FR12**	System can generate booking and revenue reports.

QUALITY ATTRIBUTES OF THE SYSTEM

Non-Functional Requirements

1

Performance

The system should retrieve seat availability and booking records efficiently.

2

Security

Only authorized users should be able to modify booking and payment data.

3

Reliability

The system should maintain accurate and consistent booking records.

4

Usability

The application should have a simple and easy-to-use interface.

5

Scalability

The database should handle increasing movies, screenings and bookings.

6

Data Integrity

Primary keys, foreign keys and constraints should prevent invalid data such as double seat booking.

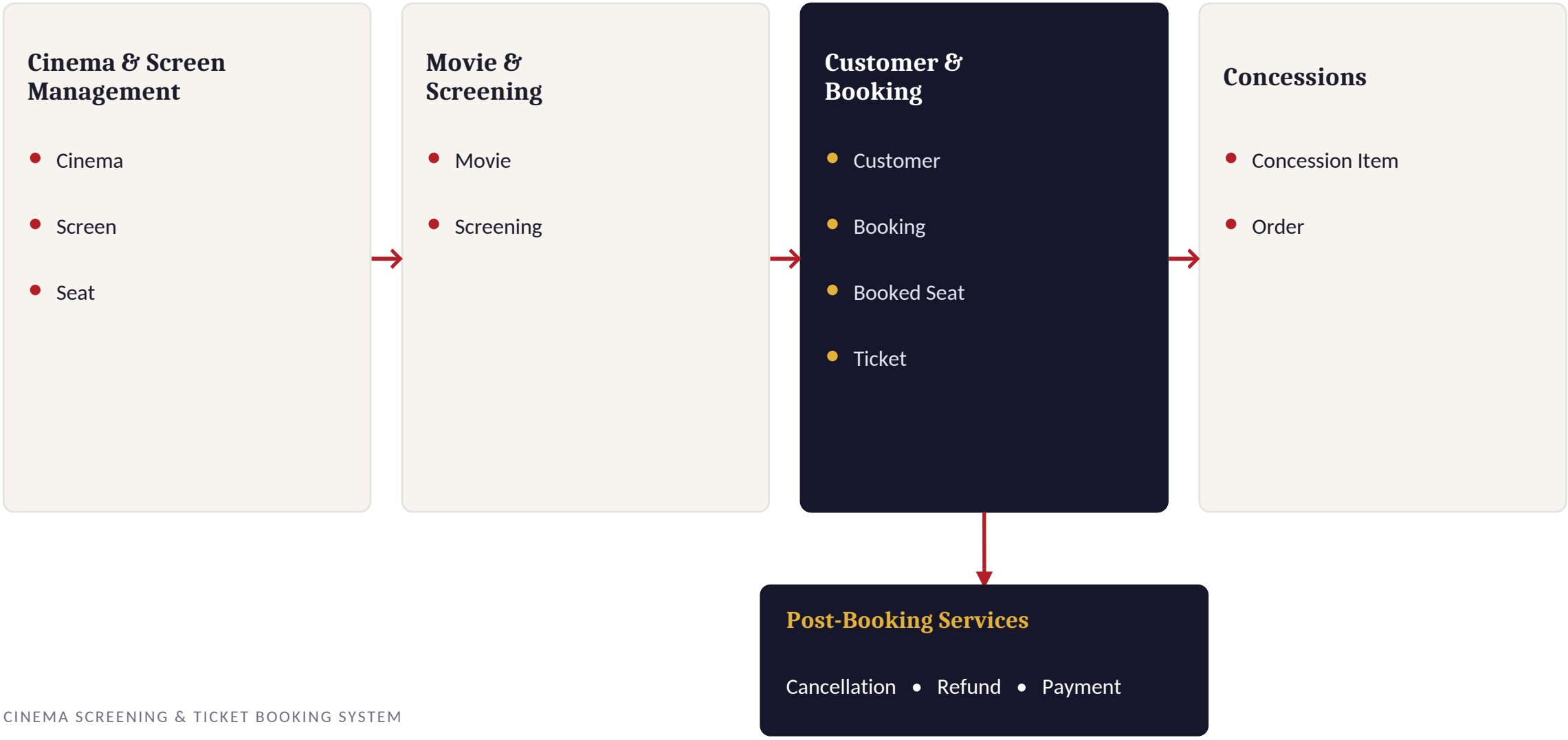
CORE ENTITIES IDENTIFIED IN THE PROJECT STATEMENT

Main Entities

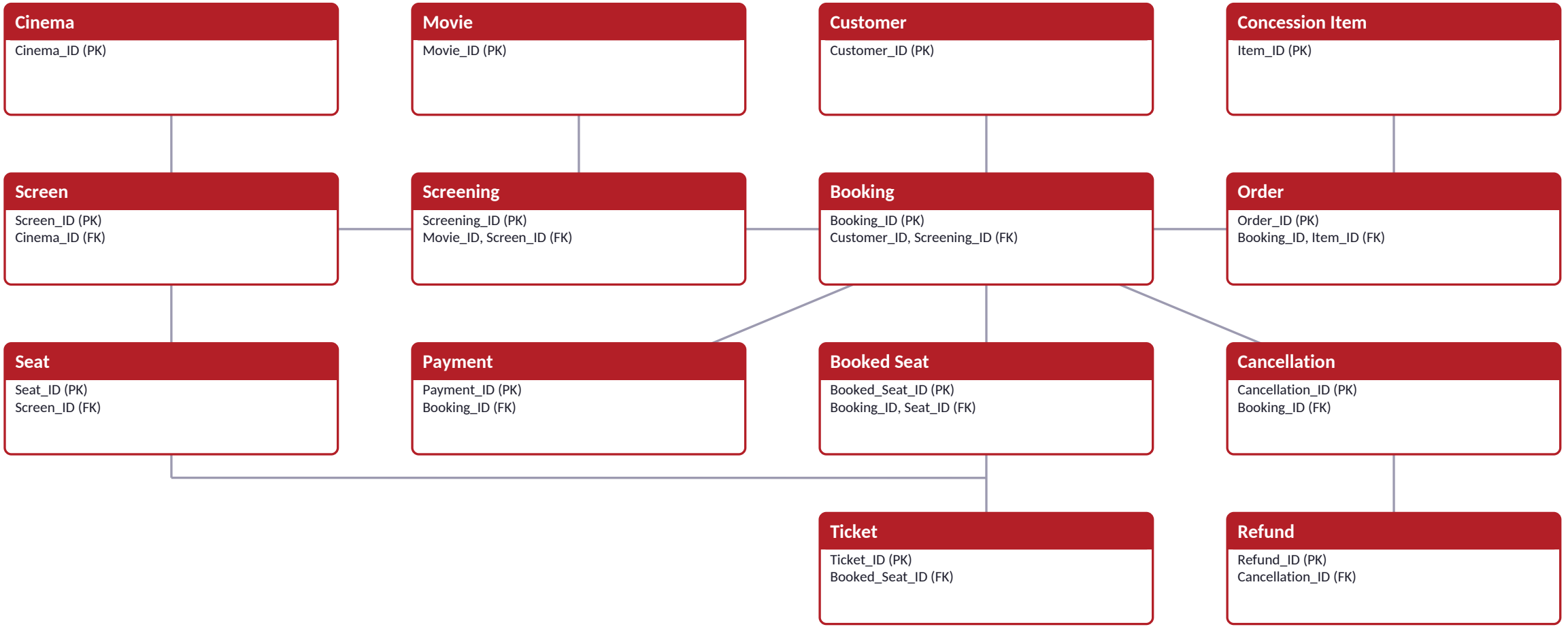
- Cinema
- Screen
- Seat
- Movie
- Screening
- Customer
- Booking
- Booked Seat
- Ticket
- Concession Item
- Order
- Cancellation
- Refund
- Payment

SIMPLIFIED CONCEPTUAL STRUCTURE

ER Diagram (Conceptual)



ER Diagram (Structured)



Line = relationship between entities (1 : N unless noted)

PROPOSED TABLES, KEYS AND RELATIONSHIPS

Initial Relational Schema

CINEMA

Cinema_ID (PK), Name, Location, Total_Screens

SCREEN

Screen_ID (PK), Cinema_ID (FK), Screen_Name, Screen_Type, Total_Seats

SEAT

Seat_ID (PK), Screen_ID (FK), Seat_Number, Seat_Type

MOVIE

Movie_ID (PK), Title, Genre, Duration, Language

SCREENING

Screening_ID (PK), Movie_ID (FK), Screen_ID (FK), Show_Date, Show_Time

CUSTOMER

Customer_ID (PK), Name, Email, Phone

BOOKING

Booking_ID (PK), Customer_ID (FK), Screening_ID (FK), Booking_Date, Status

BOOKED_SEAT

Booked_Seat_ID (PK), Booking_ID (FK), Seat_ID (FK)

TICKET

Ticket_ID (PK), Booked_Seat_ID (FK), Price, Issue_Date

CONCESSION_ITEM

Item_ID (PK), Item_Name, Price

ORDER

Order_ID (PK), Booking_ID (FK), Item_ID (FK), Quantity, Amount

CANCELLATION

Cancellation_ID (PK), Booking_ID (FK), Cancellation_Date, Reason

REFUND

Refund_ID (PK), Cancellation_ID (FK), Refund_Amount, Refund_Date

PAYMENT

Payment_ID (PK), Booking_ID (FK), Amount, Payment_Date, Payment_Method

FIVE PHASES FROM ANALYSIS TO REPORTING

Project Planning

1 Requirement Analysis

- Identify users
- Identify requirements
- Define project scope

2 Database Design

- Identify entities
- ER diagram
- Relational schema
- Normalization

3 Database Implementation

- Create tables
- Add constraints
- Insert sample data
- Functional dependencies

4 Application Development

- Develop Python app
- Implement CRUD operations
- Connect app with database
- Write SQL queries

5 Testing & Reporting

- Test database
- Test application
- Generate reports
- Prepare documentation

CONCLUSION

A centralized, structured system for cinema ticketing

The Cinema Screening and Ticket Booking System reduces manual work, improves data consistency, helps manage seat availability accurately and makes booking and payment information easier to manage and retrieve.

Thank You

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